

## **Faculty Vitae: Ezinne C. Achinivu-Ibagere**

### **1. Education**

Ph.D., Chemical and Biomolecular Engineering, North Carolina State University, 2014

M.S., Chemical and Biomolecular Engineering, North Carolina State University, 2012

B.S., Chemical Engineering, University of Maryland, College Park, 2010

### **2. Academic Experience**

Assistant Professor, University of Illinois Chicago, 2022–Present

### **3. Non-Academic Experience**

AAAS Science & Technology Policy Fellow, U.S. Department of Energy, 2020–2021

Postdoctoral Fellow, Sandia National Laboratories, 2018–2020

Guest Postdoctoral Fellow, Lawrence Berkeley National Laboratory, 2018–2020

Fulbright Postdoctoral Fellow, Chaire ABI, 2017–2018

Research Scientist/Chemist, 525 Solutions, 2015–2016

### **4. Certifications, or professional registrations**

None

### **5. Current membership in professional organizations**

### **6. Honors and Awards**

AAAS Science & Technology Policy Fellowship

Fulbright Postdoctoral Fellowship

### **7. Selected service activities**

Advisory Board Member, U.S. Department of Energy Agile BioFoundry, 2025–Present

UIC Honors College Fellow, 2023–Present

BETO BRIDGES National Review Board, 2022–2024

LSAMP / DuSable / SROP Faculty Affiliate, UIC, 2022–Present

Peer Reviewer: DOE, NSF, NIST, ACS, RSC Journals

NSBE ELEVATE National STEM Day Panelist, 2024

### **8. Selected Publications and presentations**

1. Taiwo Z. Adesanya, Jessica Fajardo, Matthew Gaughan, Ogechikanma Ihenacho, Hussein Rezk, **Ezinne Ibagere**, High-Performance Biobased Packaging: Ionic Liquid-Assisted Cellulose-Lignin Bioplastics as Sustainable Alternatives to LDPE, Chemical Engineering Journal, ***Chemical Engineering Journal***, Volume 529, 1 February 2026, 173080. <https://doi.org/10.1016/j.cej.2026.173080>.

2. Saliu, O. D.; Adesanya, T. Z.; Acquah, A.; Akobe, B.; Aganyi, D. D.; Abubakar, O. D.; **Achinivu-Ibagere, E.** Transformation of Sugar Cane Lignin into Renewable Fuel-Range Cyclo-Alkanes: In-Situ Hydrogen Release Using Earth-Abundant AlCl<sub>3</sub> and Biobased Mannose Triflate Support. *Ind. Eng. Chem. Res.* **2025**, *64* (1), 179–189. <https://doi.org/10.1021/acs.iecr.4c03341>.
3. **Ezinne C Achinivu-Ibagere**, Taiwo Zacchaeus Adesanya, Valentin Larzilliere, Erika Clavijo Rivera, Florent Allais. Charge mediated nano-filtration for recovering sinapic acid from a mustard bran hydrolysate, *Journal of Membrane Science*, 2025, 713, 123333; <https://doi.org/10.1016/j.memsci.2024.123333>.
4. **Ezinne C Achinivu**, Brian W Blankenship, Nawa Raj Baral, Hemant Choudhary, Ramu Kakumanu, Mood Mohan, Edward EK Baidoo, Corinne D Scown, Anthe George, Blake A Simmons, John Gladden. Biomass pretreatment with distillable ionic liquids for an effective recycling and recovery approach, *Chemical Engineering Journal*, 2023, 479, 147824; <https://doi.org/10.1016/j.cej.2023.147824>.
5. Paul Wolski, Brian W Blankenship, Athiyya Umar, Mica Cabrera, Blake A Simmons, Kenneth L Sale, **Ezinne C Achinivu**. Factors that influence the activity of biomass-degrading enzymes in the presence of ionic liquids—a review, *Front. Energy Res.*, 2023, 11, 1212719; <https://doi.org/10.3389/fenrg.2023.1212719>
6. **Ezinne C Achinivu**, Mica Cabrera, Athiyya Umar, Minliang Yang, Nawa Raj Baral, Corinne D Scown, Blake A Simmons, John M Gladden. In Situ Synthesis of Protic Ionic Liquids for Biomass Pretreatment, *ACS Sustainable Chem. Eng.*, 2022, 10, 37, 12090–12098; <https://doi.org/10.1021/acssuschemeng.2c01211>
7. **Ezinne C. Achinivu**, Julia L. Shamshina, Robin D. Rogers. Chitin extracted from various biomass sources: It's not the same, *Fluid Phase Equilibria*, 2022, 552; <https://doi.org/10.1016/j.fluid.2021.113286>
8. **Ezinne C. Achinivu**, Skye Frank, Nawa Raj Baral, Lalitendu Das, Mood Mohan, Peter Otoupal, Emara Shabir, Sean Utan, Corinne D. Scown, Blake A. Simmons and John Gladden. Alkanolamines as Dual Functional Solvents for Biomass Deconstruction and Bioenergy Production. *Green Chem.*, 2021, 23, 8611-8631; <https://doi.org/10.1039/D1GC02667D>
9. **Ezinne C. Achinivu**, Mood Mohan, Hemant Choudhary, Lalitendu Das, Kaixuan Huang, Harsha D. Magurudeniya, Venkataramana R. Pidatala, Anthe George, Blake A. Simmons, and John M. Gladden, A predictive toolset for the identification of effective lignocellulosic pretreatment solvents: a case study of solvents tailored for lignin extraction. *Green Chemistry*, 2021, 23, 7269-7289; <https://doi.org/10.1039/D1GC01186C>
10. Lalitendu Das, **Ezinne C. Achinivu**, Carolina Araujo Barcelos, Eric Sundstrom, Bashar Amer, Edward E. K. Baidoo, Blake A. Simmons, Ning Sun and John M. Gladden. Deconstruction of Woody Biomass via Protic and Aprotic Ionic Liquid Pretreatment for Ethanol Production. *ACS Sustainable Chem. Eng.* 2021, 9, 4422–4432. <https://doi.org/10.1021/acssuschemeng.0c07925>
11. Carolina A. Barcelos, Asun M. Oka, Jipeng Yan, Lalitendu Das, **Ezinne C. Achinivu**, Harsha Magurudeniya, Jie Dong, Simay Akdemir, Nawa Raj Baral, Chunsheng Yan, Corinne D. Scown, Deepti Tanjore, Ning Sun, Blake A. Simmons, John Gladden, and Eric

Sundstrom, High-Efficiency Conversion of Ionic Liquid-Pretreated Woody Biomass to Ethanol at the Pilot Scale, *ACS Sustainable Chem. Eng.* 2021, 9, 4042–4053; <https://doi.org/10.1021/acssuschemeng.0c07920>

12. **Ezinne C. Achinivu**, Amandine L. Flourat, Fanny Brunissen, Florent Allais. Valorization of waste biomass from oleaginous “oil-bearing” seeds through the biocatalytic production of sinapic acid from mustard bran. *Biomass and Bioenergy*, 2021, 145; <https://doi.org/10.1016/j.biombioe.2020.105940>.
13. **Ezinne C. Achinivu**, Protic Ionic Liquids for Lignin Extraction—A Lignin Characterization Study. *International Journal of Molecular Sciences*, 2018, 19, 428; <https://doi.org/10.3390/ijms19020428>.
14. **Ezinne C. Achinivu**, Reagan M. Howard, Guoqing Li, Hanna Gracz and Wesley A. Henderson, Lignin extraction from biomass with protic ionic liquids. *Green Chemistry*, 2014, 16, 1114-1119; <https://doi.org/10.1039/C3GC42306A>

#### **Relevant Intellectual Property**

- US Appl 17/547,134; Dec 9, 2021; Use of polyamines in the pretreatment of biomass.
- Provisional; US 63/035,508; Jun 5, 2020; Use of "in-situ" ionic liquid (IL) and deep eutectic solvent (DES) synthesis using chemically synthesized or biomass-derived ions in the pretreatment of biomass.
- Provisional; US 63/023,770; May 12, 2020; Use of alkanolamines for lignin extraction in the pretreatment of biomass.
- Provisional; US 62/978,771; Feb 19, 2020; Use of distillable volatile salt for the pretreatment of biomass.

#### **Presentations**

- **Achinivu, E. C.**, et al *ACS National Meeting*, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Spring 2024
- **Achinivu, E. C.**, et al *AIChE Annual Meeting*, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025
- **Achinivu, E. C.**, et al *Symposium on Biomaterials, Fuels and Chemicals* 2024, 2025 (Session Chair)
- **Achinivu, E. C.**, et al *Congress of Ionic Liquids (COIL)*, March 2019

#### 9. **Professional Development Activities**

Not applicable